PROBLEM CL12

# Gas Discharging through an Inverted U-Pipe: Outlet State, Force and Moment

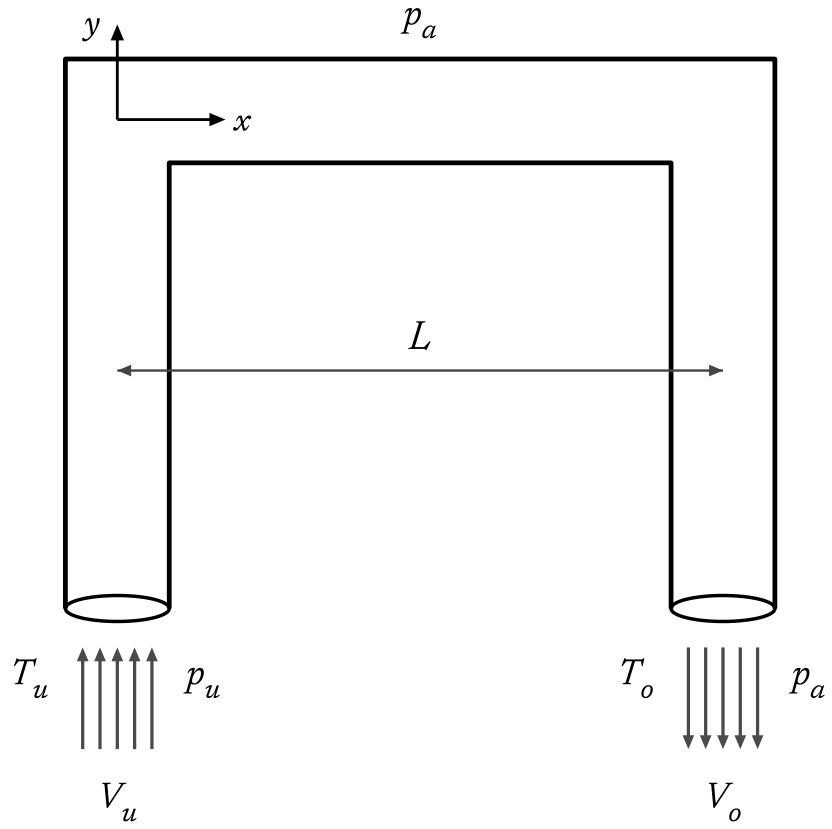
Original statement · CL12.pdf

Compressible | Energy | Linear momentum | Angular momentum

## OFFICIAL PROBLEM STATEMENT

The gas circulating along the pipe of cross section $A$ depicted in the figure of CL12.pdf discharges to the atmosphere, where the pressure is $p_a$. Measurements at a given upstream location give $V_u$, $p_u$, and $T_u$ for the velocity, pressure and temperature, all uniform across the section. In the analysis that follows, it is assumed that the pipe is thermally insulated and that buoyancy forces have a negligible effect on the gas motion. The gas properties are defined by its gas constant $R_g$ together with the ratio of specific heats $\gamma = c_p/c_v$.

- Obtain the values of the gas temperature and velocity at the outlet section $T_o$ and $V_o$ in terms of $V_u, p_u, T_u, p_a, R_g$ and $\gamma$.
- Compute the two components of the force $F_x$ and $F_y$ acting on the stretch of pipe considered.
- Calculate the momentum on the pipe with respect to the origin of the coordinate system indicated in the figure.




Statement figure (redrawn from CL12.pdf, p. 1): inverted U-pipe of section $A$; inlet state $T_u$, $p_u$, $V_u$ in the left leg, outlet $T_o$, $p_a$, $V_o$ in the right leg, legs $L$ apart; origin on the axis of the left leg

Source: CL12.pdf (page 1, exam of Dec 16, 2011). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral angular momentum balance Eq. 3.38, integral total energy balance Eq. 3.46.

**Worked solution (4-phase method)**

PHASE 1

## Hypotheses, Data and Unknowns

- Steady compressible flow of a perfect gas ($p = \rho R_g T$, $h = c_p T = \frac{\gamma}{\gamma-1} R_g T$); neglect $\bar{g}$; pipe insulated and at rest (no heat, no work); uniform properties at both sections.
- Geometry (figure): inverted U of constant section $A$; the gas goes up the left leg ($\bar{v} = V_u \bar{e}_y$ at the measuring section) and down the right leg, discharging as a free jet ($p = p_a$, $\bar{v} = -V_o \bar{e}_y$) at a horizontal distance $L$ from the left leg; the origin $O$ is on the axis of the left leg.

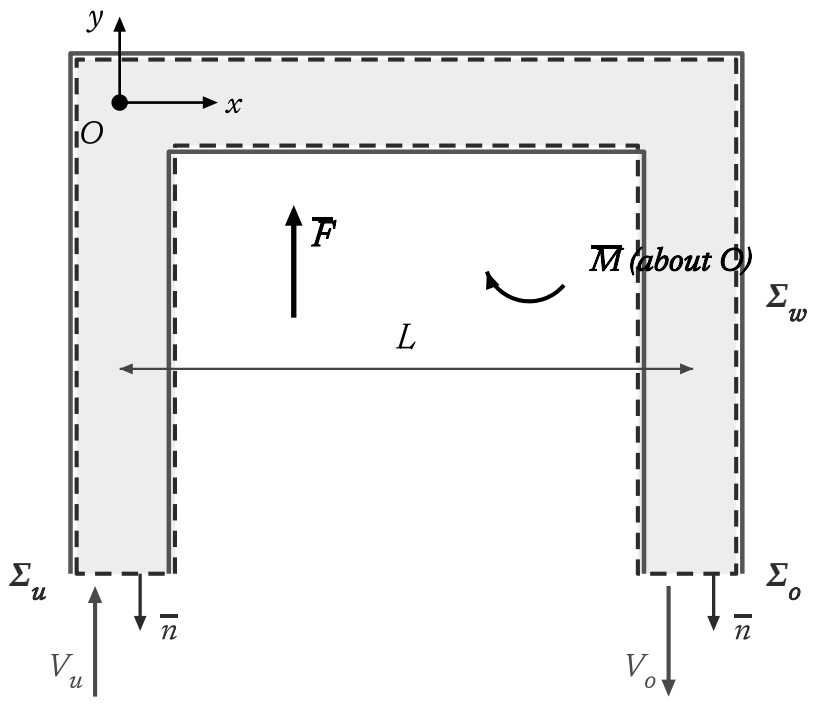
- Data: $[V_u,\ p_u,\ T_u,\ p_a,\ R_g,\ \gamma,\ A,\ L]$. Unknowns: $[T_o,\ V_o,\ F_x,\ F_y,\ \bar{M}]$. The 'momentum on the pipe' of the statement is the moment of the forces on the pipe about $O$.

| Symbol | Meaning | SI unit |
|---|---|---|
| $V_u, p_u, T_u$ | upstream state | m/s, Pa, K |
| $V_o, T_o$ | outlet velocity and temperature | m/s, K |
| $A, L$ | pipe section, distance between legs | $m^2$, m |
| $\bar{F}, \bar{M}$ | force and moment on the pipe | N, N·m |

PHASE 2

## Frames, Control Volume and Surfaces

- Frame fixed to the pipe; $\bar{e}_x$ horizontal (from the left leg to the right leg), $\bar{e}_y$ vertical upwards, $\bar{e}_z = \bar{e}_x \wedge \bar{e}_y$ towards the reader. Fixed control volume.
- Surfaces: $\Sigma_u$ (measuring section, $\bar{n} = -\bar{e}_y$), $\Sigma_o$ (outlet, $\bar{n} = -\bar{e}_y$), $\Sigma_w$ (inner wall). Positions: $\bar{x} = y\bar{e}_y + \bar{\eta}$ on $\Sigma_u$ and $\bar{x} = L\bar{e}_x + y\bar{e}_y + \bar{\eta}$ on $\Sigma_o$, with $\int \bar{\eta}\, d\sigma = 0$.




Control volume of the solution: the gas in the pipe, with $\Sigma_u$ (measuring section) and $\Sigma_o$ (outlet), both with $\bar{n} = -\bar{e}_y$, and the wall $\Sigma_w$; moments about $O$ on the axis of the left leg. $\bar{F}$ points up and $\bar{M}$ is counter-clockwise ($+\bar{e}_z$), as in the results

$$\boxed{\bar{F} = \int_{\Sigma_w} \left[ (p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

$$\boxed{\bar{M} = \int_{\Sigma_w} \bar{x} \wedge \left[ (p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

PHASE 3

## Balances, Surface by Surface

Mass (Notes Eq. 3.11)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho \, dV}}_{0 \text{ (steady)}} + \int_{\Sigma_c} \rho \left( \bar{v} - \underbrace{\cancel{\bar{v}_c}}_{0} \right) \cdot \bar{n} \, d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

| **Surface** | $\int \rho \, \bar{v} \cdot \bar{n} \, d\sigma$ |
|---|---|
| $\Sigma_u$ | $-\rho_u V_u A$, with $\rho_u = \frac{p_u}{R_g T_u}$ |
| $\Sigma_o$ | $+\rho_o V_o A$, with $\rho_o = \frac{p_a}{R_g T_o}$ |
| $\Sigma_w$ | $0$ |

$$\rho_u V_u A = \rho_o V_o A \implies \boxed{\frac{p_u V_u}{T_u} = \frac{p_a V_o}{T_o}} \qquad (1)$$

Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho\left(e+\frac{v^2}{2}\right)dV}_{0\ \text{(steady)}}+\int_{\Sigma_c}\rho\left(e+\frac{v^2}{2}\right)\left[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}\right]d\sigma=-\int_{\Sigma_c}p\,\bar{v}\cdot\bar{n}\,d\sigma+\int_{\Sigma_c}\bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma+\underbrace{\int_{V_c}\rho\bar{g}\cdot\bar{v}\,dV}_{\text{neglect }\bar{g}}-\int_{\Sigma_c}\bar{q}\cdot\bar{n}\,d\sigma+\underbrace{\int_{V_c}Q_R\,dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

| **Surface** | $\int\rho\left(e+\frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma$ | $-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$ | $\int\bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$ | $-\int\bar{q}\cdot\bar{n}\,d\sigma$ |
|---|---|---|---|---|
| $\Sigma_u$ | $-\dot{m}\left(e_u+\frac{V_u^2}{2}\right)$ | $+p_uV_uA$ | $0$ | $0$ |
| $\Sigma_o$ | $+\dot{m}\left(e_o+\frac{V_o^2}{2}\right)$ | $-p_aV_oA$ | $0$ | $0$ |
| $\Sigma_w$ | $0$ | $0$ ($\bar{v}=0$) | $0$ ($\bar{v}=0$) | $0$ (insulated) |

$$\boxed{\frac{\gamma}{\gamma-1}R_gT_u+\frac{1}{2}V_u^2=\frac{\gamma}{\gamma-1}R_gT_o+\frac{1}{2}V_o^2}\qquad(2)$$

Solving (1) and (2): with $C\equiv\frac{p_uV_u}{p_aT_u}$, (1) gives $V_o=C\,T_o$; substituting into (2), $\frac{C^2}{2}T_o^2+\frac{\gamma R_g}{\gamma-1}T_o-H=0$ with $H=\frac{\gamma R_g}{\gamma-1}T_u+\frac{V_u^2}{2}$, whose positive root is

$$\boxed{T_o=\frac{-\frac{\gamma R_g}{\gamma-1}+\sqrt{\left(\frac{\gamma R_g}{\gamma-1}\right)^2+2C^2H}}{C^2},\qquad V_o=C\,T_o}\qquad(3)$$

`Linear momentum (Notes Eq. 3.36)`

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}_{0\ \text{(steady)}}+\int_{\Sigma_c}\rho\bar{v}\left[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}\right]d\sigma=-\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma+\int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma+\underbrace{\int_{V_c}\rho\bar{g}\,dV}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

| **Surface** | $\int\rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$ | $-\int(p-p_a)\,\bar{n}\,d\sigma$ | $\int\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$ |
|---|---|---|---|
| $\Sigma_u$ | $\rho_uV_u\bar{e}_y(-V_u)A=-\rho_uV_u^2A\,\bar{e}_y$ | $-(p_u-p_a)(-\bar{e}_y)A=(p_u-p_a)A\,\bar{e}_y$ | $0$ |

| **Surface** | $\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$ | $-\int(p-p_a)\,\bar{n}\,d\sigma$ | $\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$ |
|---|---|---|---|
| $\Sigma_o$ | $\rho_o(-V_o\bar{e}_y)(V_o)A = -\rho_u V_u A\,V_o\,\bar{e}_y$ | 0 (free jet) | 0 |
| $\Sigma_w$ | 0 | $-\int_{\Sigma_w}(p-p_a)\,\bar{n}\,d\sigma$ | $\int_{\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$ |

Boxed $\Sigma_w$ terms: $-\bar{F}$. The $\Sigma_o$ entry uses (1).

$$-\rho_u V_u A\,(V_u+V_o)\,\bar{e}_y = (p_u-p_a)A\,\bar{e}_y - \bar{F} \implies \boxed{F_x = 0, \qquad F_y = \rho_u V_u A\,(V_u+V_o)+(p_u-p_a)A} \quad (4)$$

**Angular momentum about O (Notes Eq. 3.38)**

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\,\bar{x}\wedge\bar{v}\,dV}}_{0} + \int_{\Sigma_c}\rho\,(\bar{x}\wedge\bar{v})(\bar{v}\cdot\bar{n})\,d\sigma = -\int_{\Sigma_c}\bar{x}\wedge(p-p_a)\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

| **Surface** | $\int\rho\,(\bar{x}\wedge\bar{v})(\bar{v}\cdot\bar{n})\,d\sigma$ | $-\int\bar{x}\wedge(p-p_a)\bar{n}\,d\sigma$ | $\int\bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$ |
|---|---|---|---|
| $\Sigma_u$ | 0 ($y\bar{e}_y\wedge\bar{e}_y=0$, $\int\bar{\eta}\,d\sigma=0$) | 0 (same reason) | 0 |
| $\Sigma_o$ | $\rho_o V_o A\,(L\bar{e}_x\wedge(-V_o\bar{e}_y)) = -\rho_u V_u A\,LV_o\,\bar{e}_z$ | 0 (free jet) | 0 |
| $\Sigma_w$ | 0 | $-\int_{\Sigma_w}\bar{x}\wedge(p-p_a)\bar{n}\,d\sigma$ | $\int_{\Sigma_w}\bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$ |

Boxed $\Sigma_w$ terms: $-\bar{M}$.

$$-\rho_u V_u A\,LV_o\,\bar{e}_z = -\bar{M} \implies \boxed{\bar{M} = \rho_u V_u A\,L\,V_o\,\bar{e}_z = \frac{p_u}{R_g T_u}V_u A\,L\,V_o\,\bar{e}_z} \quad (5)$$

PHASE 4

## Interpretation, Limits and Dimensional Check

- Adiabatic, workless flow keeps the stagnation enthalpy (2); the pressure drop to $p_a$ lowers the density, so the gas accelerates ($V_o > V_u$) and cools slightly ($T_o < T_u$).

- Both legs push the pipe upwards: the gas turning down in the right leg and the gauge pressure at the measuring section. The moment comes only from the outlet jet, offset by $L$ from $O$.
- Incompressible-like limit ($p_u \to p_a$, small $V$): $T_o \to T_u$, $V_o \to V_u$, $F_y \to 2\rho_u V_u^2 A$.
- Dimensions (SI): $[\rho_u V_u^2 A] = \mathrm{N}$; $[\rho_u V_u A L V_o] = \mathrm{N \cdot m}$; $[R_g T] = [V^2] = \mathrm{m^2/s^2}$.
- Numerical illustration (hypothetical data, air): $V_u = 50$ m/s, $p_u = 1.5 \times 10^5$ Pa, $T_u = 400$ K, $p_a = 101325$ Pa, $R_g = 287$ J/(kg K), $\gamma = 1.4$, $A = 0.01$ m$^2$, $L = 1$ m give $T_o = 398.537$ K, $V_o = 73.749$ m/s, $F_y = 567.6$ N, $M_z = 48.181$ N·m (residuals of (1), (2) below $10^{-8}$).
- Agreement with the official solution: CL12.pdf obtains (1), (2), (4) and (5); it does not solve (1)-(2) explicitly.